10,000 URL evaluation

9,700 legitimate URLs • 300 phishing URLs • unchanged thresholds

Condition	Accuracy	Precision	Recall	F1	ROC-AUC	PR-AUC	FPR	FNR
HYB-MS real-only	98.71%	77.14%	81.00%	79.02%	96.50%	85.27%	0.74%	19.00%
HYB-MS +10% GAN	98.80%	78.48%	82.67%	80.52%	96.61%	87.11%	0.70%	17.33%
Browser candidate	98.80%	78.48%	82.67%	80.52%	96.61%	87.10%	0.70%	17.33%
Extension v0.5.1	98.91%	85.77%	76.33%	80.78%	96.61%	87.10%	0.39%	23.67%

GAN recall: 82.67% | GAN FNR: 17.33%

GAN minus real-only F1: +1.50 points

Extension v0.5.1 recall: 76.33% | FNR: 23.67%

The cohort is identity-disjoint from prior project URL files. Tranco roots are legitimate proxies; PhishTank rows were verified online in the cached source snapshot.

10,000 URL Evaluation — 300 Phishing URLs















